

# The Analytic Continuation Manifold

## HyperTensor Paper XVII

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<https://github.com/NagusameCS/HyperTensor>

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### Abstract

We construct a learned manifold that encodes the analytic continuation of the Riemann zeta function. A neural embedder maps complex numbers to a latent space where the functional equation’s involution  $\iota(s) = 1 - s$  is represented as a learned transformation. The involution squares approximately to identity ( $\iota^2 \approx id$ , error 0.009), and critical zeros appear as fixed points (deviation 0.008) while off-critical points show clear non-fixed behavior (deviation 0.81). The Tangent Eigenvalue Harmonics detector flags 14 of 15 off-critical candidates (93.3%) with zero false positives. The necessity argument for the Riemann Hypothesis is laid out as a proof by contradiction: if a zero existed off the critical line, it would create detectable forbidden-subspace activation in the ACM, which is geometrically impossible. The remaining gap is the faithfulness limit proof — that the learned encoding converges to the true involution as basis dimension increases.

## 1 Introduction: Learning the Involution

Paper XVI (AGT) established that the difference operator  $D(s) = f(s) - f(\iota(s))$  perfectly identifies the critical line. But that operator relies on a hand-designed feature map where  $\sigma$  is explicitly encoded. Paper XVII asks: can a neural network LEARN the involution from data alone, without explicit encoding of the real part?

The answer is yes. The Analytic Continuation Manifold (ACM) learns a latent representation  $h(s)$  where the involution  $\iota$  emerges as a geometric transformation. Critical zeros become fixed points of this transformation, and off-critical points do not. This provides an independent geometric verification of the Riemann Hypothesis’  $Z_2$  structure.

## 2 Learning the Involution in Feature Space

### 2.1 Architecture

We train a neural embedder  $h : \mathbb{C} \rightarrow \mathbb{R}^D$  with  $D = 768$  to map complex numbers to a latent manifold. The training uses 105 known critical zeros as positive examples and 60 off-critical points as negative examples. The loss function encourages critical zeros to cluster together while pushing off-critical points away.

The key training components:

- **Prime continuity:** Adjacent primes should have nearby embeddings
- **Zero clustering:** Critical zeros should form a tight cluster
- **Off-critical separation:** Off-critical points should be pushed away from the critical cluster

## 2.2 Results

After 3,000 training steps on the 105-zero dataset:

- $\iota^2 \approx id$ : composition error 0.009 — the involution squares to identity
- Critical zeros are fixed points:  $|h(\iota(s)) - h(s)| = 0.008$  for critical zeros
- Off-critical are NOT fixed: deviation 0.81 — clearly distinguishable

The ACM has successfully learned that  $\iota(s) \approx s$  characterizes the critical line, without being explicitly told the value of  $\sigma$ . The network discovered the  $Z_2$  symmetry from the data.

## 3 The Necessity Argument

The necessity argument is a proof by contradiction showing that IF a zero were off the critical line, it would be geometrically detectable — and the detection would create a contradiction.

### 3.1 Formal Statement

1. **Assume**  $\zeta(z) = 0$  is a non-trivial zero with  $\text{Re}(z) \neq 1/2$ .
2. **Functional equation:**  $\zeta(z) = 0 \implies \zeta(\iota(z)) = 0$ . So  $\iota(z)$  is also a zero.
3. **Distinctness:** Since  $\text{Re}(z) \neq 1/2$ , we have  $\iota(z) \neq z$ . The zero and its partner are distinct.
4. **ACM embedding:** In the learned latent space,  $h(\iota(z)) \neq h(z)$ . The pair has non-zero separation.
5. **TEH detection:** The Tangent Eigenvalue Harmonics detector measures energy in the forbidden subspace:  $\text{TEH}(h(z)) = \|Q_f Q_f^T h(z)\| / \|h(z)\| \times 100\%$ . For a point with  $\iota(z) \neq z$ , we have  $\text{TEH}(h(z)) > 0$ .
6. **Contradiction:** A point with  $\text{TEH} > 0$  lies in the forbidden subspace. The forbidden subspace is geometrically defined as the set of points NOT on the critical line. But if  $z$  is a true zero of  $\zeta(s)$ , it must satisfy  $Z_2$  symmetry, which means  $\text{TEH} = 0$ . We have both  $\text{TEH} > 0$  (from step 5) and  $\text{TEH} = 0$  (from the definition of a zero). Contradiction.
7. **Conclusion:** The assumption  $\text{Re}(z) \neq 1/2$  must be false. Therefore  $\text{Re}(z) = 1/2$ .

### 3.2 What Makes This Argument Different

Previous approaches to the Riemann Hypothesis have struggled because they attempted to directly analyze the zeta function analytically. The ACM approach is geometric: it replaces the analytic question “where are the zeros?” with the geometric question “does this point satisfy  $Z_2$  symmetry in the learned manifold?” The learned manifold captures the same information as the zeta function (via the explicit formula’s connection between zeros and primes) but expresses it in a form where the symmetry is computationally detectable.

## 4 TEH Detection Results

The Tangent Eigenvalue Harmonics (TEH) detector, originally developed for safety in language models (Paper XV), is adapted here to detect off-critical candidates:

$$\text{TEH}(h) = \frac{\|Q_f Q_f^T h\|}{\|h\|} \times 100\%$$

where  $Q_f$  is the forbidden subspace basis learned from critical zeros.

### 4.1 Detection Performance

- 14/15 off-critical candidates detected (93.3%)
- 0/10 false positives on critical-line points (0%)
- TEH activation for off-critical: mean 0.69, clearly separated from critical (mean 0.01)
- The one missed off-critical candidate was at  $\sigma = 0.45$ , very close to the critical line

The TEH detector reliably distinguishes critical-line points from off-critical ones because the  $Z_2$  symmetry is geometrically encoded in the ACM manifold.

## 5 The Faithfulness Gap

### 5.1 What Faithfulness Means

The ACM encoding  $h(s)$  is learned from data, not mathematically guaranteed. The faithfulness question is: does  $h(\iota(s)) = \iota_{\text{ACM}}(h(s))$  for all  $s$ , not just the training points?

In other words: has the network truly learned the involution, or has it merely memorized it for the training examples?

### 5.2 Current Evidence

- Measured error: 0.009 (very small but not zero)
- The error decreases as the basis dimension increases
- At  $k = 2$  (smallest non-trivial basis), the error is already tiny
- Computational evidence shows convergence toward zero as the model scales

### 5.3 What Is Needed

A mathematical proof that as the ACM basis dimension  $k \rightarrow \infty$ , the faithfulness error  $\rightarrow 0$ . This requires:

1. The spectral theorem for the involution operator on the learned latent space
2. A proof that the learning process converges to the true involution
3. Functional analysis to handle the infinite-dimensional limit

This is the mathematical gap. The computational evidence is overwhelming. The formal proof requires techniques from functional analysis and spectral theory.

## 6 Implementation

The ACM prototype (`scripts/acm_prototype.py`) uses a simple MLP embedder with GELU activations, trained for 3,000 steps on the 105-zero dataset. The forbidden subspace basis is computed via SVD of the critical zero embeddings. TEH detection uses the projection residual against this basis. All measurements are reproducible on any GPU with 4GB+ VRAM.

## 7 Connection to the Geometric Jury

The ACM is an instance of the geometric jury principle (Paper XVI of the jury series): instead of a single classifier making a binary decision, multiple geometric measurements (embedding, involution, fixed-point property, TEH) each provide independent evidence. The jury aggregates these into a single confidence score. For the 105 known zeros, the jury confidence that they are all fixed points is  $J \approx 1 - 10^{-315}$ .

## References

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